

NONPARAMETRIC CONFIDENCE SEQUENCES FOR THE CENTRE OF SYMMETRY OF A SYMMETRIC DISTRIBUTION

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1. Introduction

A $1-2\alpha$ confidence sequence for an unknown parameter θ is a sequence of random intervals $\{I_n\}$ such that $P_\theta(\theta \in I_n \text{ for all } n \geq n_0) \geq 1-2\alpha$ for all θ and some positive integer n_0 . Wald (1947) briefly touched upon confidence sequences (although he did not use this terminology) in sequential estimation. Robbins (1970) discussed the construction and some applications of confidence sequences. For example, if I_n depends on the first n observations in sequential sampling, the sequence may be constructed sequentially and stopped at an arbitrary (even random) point N . The resulting interval I_N still contains θ with probability at least $1-2\alpha$. In particular, N may be the first integer $n \geq n_0$ such that I_n is of length not larger than a given number. In this way a bounded length confidence interval procedure is obtained. Confidence sequences may be used in the construction of sequential tests with power one and sequential selection procedures.

Bounded length sequential confidence intervals for the point of symmetry of a distribution based on one-sample rank tests were proposed and studied by Sen and Ghosh (1971), following earlier work by Chow and Robbins (1965) (using intervals based on the sample mean) and Geertsema (1970) (who used intervals based on the sign test and the Wilcoxon one-sample test).

We show how confidence sequences based on one-sample rank tests may be constructed for the centre of symmetry of a distribution and study the associated bounded length confidence interval procedures. In the asymptotic analysis of the latter procedures (in two limiting situations) we make use of the results of Sen and Ghosh (1971) and usually follow their notation. We remark that the main difference between their procedures and ours is that theirs attain a prescribed probability of coverage only in an asymptotic sense, while our procedure guarantees a minimum coverage probability $1-2\alpha$. On the other hand the asymptotic expected sample sizes of our procedures are of slightly higher order of magnitude.

2. A method for the construction of confidence sequences

Denote by $X_1, X_2, \dots, X_n, \dots$ a sequence of iid random variables such that the common distribution function $F(x-\theta)$ is symmetric about a point θ .

We shall base confidence sequences for θ on sequences of statistics $\{T_n(X_n)\}$ where $X_n = (X_1, \dots, X_n)$ and where it is assumed that $T_n(X_n - d\mathbf{1}_n) = T_n(X_1 - d, \dots, X_n - d)$ is a nonincreasing function of d . Here $\mathbf{1}_n = (1, 1, \dots, 1)$. Let $S_n = n(T_n - \nu)$, where $T_n = T_n(X_n)$, $\nu = ET_n$ when $\theta = 0$, so that $ES_n = 0$ when $\theta = 0$. In the cases to be considered, T_n will be a test statistic for testing the hypothesis of symmetry about 0 of the underlying distribution. We will show how, for prescribed $\alpha (0 < \alpha < 1)$ and for a positive integer n_0 , a sequence of positive constants $\{c(n, \alpha)\}$ may be constructed such that when $\theta = 0$

$$(1) \quad P(S_n \geq c(n, \alpha) \text{ for some } n \geq n_0) \leq \alpha \text{ and}$$

$$P(S_n \leq -c(n, \alpha) \text{ for some } n \geq n_0) \leq \alpha.$$

Now let

$$(2) \quad V_n = \inf \{d : S_n(X_n - d\mathbf{1}_n) \leq c(n, \alpha)\}$$

$$U_n = \sup \{d : S_n(X_n - d\mathbf{1}_n) \geq -c(n, \alpha)\}.$$

(If $\sup S_n(X_n - d\mathbf{1}_n) \leq c(n, \alpha)$, the inequality of the definition of V_n is always satisfied and thus $V_n = -\infty$. Similarly we may obtain $U_n = +\infty$.)

For the cases considered in this article it may be shown that $V_n < U_n$ for all $n \geq n_0$ where n_0 is a nonrandom positive integer. It follows directly from (1) and (2) that $\{(V_n, U_n)\}$ is a $1-2\alpha$ confidence sequence for θ . It may be noted that V_n and U_n are translation invariant.

We remark that the above is analogous to the usual method for constructing confidence intervals based on nonparametric tests.

3. Confidence sequences based on one-sample rank statistics

As indicated in Section 2 we now derive confidence sequences from the one-sample rank statistics

$$(3) \quad T_n = n^{-1} \sum_{i=1}^n J_n(i/(n+1)) Z_{i(n)}$$

where $Z_{i(n)} = 1$ if the X_j corresponding to the i -th smallest under the absolute X_j 's is positive and $Z_{i(n)} = 0$ otherwise. We assume that $\{J_n(u) : 0 < u < 1\}$ is generated by the score-function $\{J(u) : 0 \leq u < 1\}$ as follows: $J_n(u) = EJ(Y_{ni})$; $(i-1)/n < u \leq i/n$; $1 \leq i \leq n$, where $Y_{n1} \leq \dots \leq Y_{nn}$ are the order statistics of a sample from a distribution uniform on $(0, 1)$. Let J and F have the properties assumed in section 2 of Sen and Ghosh

(1971). Also let $A_n^2 = n^{-1} \sum_{i=1}^n J_n^2(i/(n+1))$, $\mu = \int_0^1 J(u) du$ and $A^2 = \int_0^1 J^2(u) du$.

Then $A_n^2 \uparrow A^2$.

If we define V_n and U_n as in (2), then it follows from their translation invariance that we may assume, without loss of generality, that $\theta = 0$. This assumption will be made in all the proofs to follow. In that case it follows from Hajek (1969) Theorem 19C, that $Z_{i(n)}$ are iid with $P(Z_{i(n)} = 1) = 1/2$ for $i=1, \dots, n$. Hence $ET_n = 1/2\mu$. Also,

$$\begin{aligned} E \exp(aS_n) &= \prod_{i=1}^n \frac{1}{2} \{ \exp(\frac{1}{2} a J_n(i/(n+1))) + \exp(-\frac{1}{2} a J_n(i/(n+1))) \} \\ &\leq \exp(a^2 n A_n^2 / 8) \leq \exp(a^2 n A^2 / 8) \end{aligned}$$

According to Sen and Ghosh (1971), Theorem 4.5 the sequence $\{S_n\}$ is a martingale sequence where S_n is defined in Section 2 and T_n is given by (3).

Applying Theorem A of the Appendix we see that (1) is satisfied by

$$(4) \quad c(n, \alpha) = \frac{1}{2} B A (\gamma n - 1)^{1/2} (\log \log n - b \log \alpha)^{1/2}$$

with a suitable choice of an integer $\gamma \geq 2$ and constants $b > 0$ and $B > 2^{1/2}$.

This follows from the fact that

$$\begin{aligned} P(S_n \geq c(n, \alpha) \text{ for some } n \geq m) \\ \leq \alpha^{1/2} B^2 b^{1/2} (\frac{1}{2} B^2 - 1 + \log m / \log \gamma) / (\frac{1}{2} B^2 - 1) (\log m)^{1/2} B^2 \\ = \alpha \text{ for such a choice of } \gamma, b \text{ and } B. \end{aligned}$$

In what follows we shall let γ , B and m be constants and $b = b(\alpha)$ be a function of α . Then

$$(5) \quad b = (2/B^2) \{1 + [\frac{1}{2} B^2 \log \log m + \log((\frac{1}{2} B^2 - 1)/(\frac{1}{2} B^2 - 1 + \log m / \log \gamma))]/\log \alpha\}.$$

We remark that Sen and Ghosh (1973) have proved a Law of the iterated logarithm for one-sample rank statistics. This shows that a sequence of the order $(n \log \log n)^{1/2}$ such as our sequence (4) is in a sense the best possible (compare with Robbins (1970)).

4. Asymptotic properties of a related bounded length confidence interval procedure

Define the stopping variable N as the first integer $n \geq m$ such that $U_n - V_n \leq L$ where $L > 0$. Then (V_N, U_N) is a $1-2\alpha$ confidence interval for the point of symmetry of the underlying distribution with length at most L .

In order to investigate the asymptotic expected sample size, we need the following result which is analogous to the asymptotic expansion of Bahadur (1966) for sample quantiles:

Lemma 1

$$(6) \quad V_n = \theta + (S_n - c(n, \alpha)) / nD(F) + R_n$$

$$U_n = \theta + (S_n + c(n, \alpha)) / nD(F) + R_n$$

where $D(F) = \int_0^\infty (d/dx) J[2F(x) - 1] dF(x)$,

and where R_n is generically used to denote a random variable such that for given $s > 0$ constants K_s and K'_s and an integer n_s may be found with

$$(7) \quad P(|R_n| > K_s n^{-3/4} (\log n)^6) \leq K'_s n^{-s}$$

for all $n \geq n_s$. Then $R_n = o(n^{-3/4} (\log n)^6)$ a.s.

Proof:

In the same way as lemma 5.1 of Sen and Ghosh (1971) was proved, we obtain: for each $s > 0$ there are constants c_s and c'_s and an integer n_s such that

$$(8) \quad P_0(|V_n| < c_s n^{-1/2} (\log n)^2) \geq 1 - c'_s n^{-s} \text{ for all } n \geq n_s.$$

From (8) and Theorem 4.3 of Sen and Ghosh now follows that for each $s > 0$ there are constants k_s and k'_s and an integer n_s such that for all $n \geq n_s$

$$(9) \quad P_0(|W_n(V_n + 1/n)| \leq k_s n^{-1/4} (\log n)^6) \geq 1 - k'_s n^{-s}$$

where

$$(10) \quad W_n(a) = n^{-1/2} (S_n(X_n) - S_n(X_n - a|_n) - nD(F)).$$

Substituting $a = V_n + 1/n$ in (10) and using the fact that

$$S_n(X_n - (V_n + 1/n)|_n) \leq c(n, \alpha)$$

we obtain

$$V_n \geq (S_n - c(n, \alpha)) / nD(F) + R_n.$$

The opposite inequality follows similarly by substituting $a = V_n - 1/n$ in (10). The expansion for U_n is proved in the same way. The last statement of the lemma follows from the Borel-Cantelli Lemma.

Corollary 1

It follows immediately from (6) that

$$(11) \quad n^{1/2}(\log \log n)^{-1/2}(U_n - V_n) \rightarrow \gamma^{1/2}BA/D(F) \text{ a.s. as } n \rightarrow \infty.$$

We first consider the asymptotic behaviour of N as $L \rightarrow 0$.

Theorem 1

- (i) $N < \infty$ for all $L > 0$ and $\alpha > 0$, $N = N(L)$ is a function of L which is nondecreasing as L decreases, $N \rightarrow \infty$ as $L \rightarrow 0$ and $EN \rightarrow \infty$ as $L \rightarrow 0$.
- (ii) $N/H(L) \rightarrow 1$ a.s. as $L \rightarrow 0$ where $H(L) = \gamma B^2 A^2 \log \log L^{-2}/L^2 D^2(F)$.
- (iii) The set $\{N/H(L)\}_{L>0}$ is uniformly integrable and $EN/H(L) \rightarrow 1$ as $L \rightarrow 0$.

Proof.

(i) and (ii) follow from Lemma 1 of Chow and Robbins (1965) together with (11).

To prove (iii), it is sufficient to prove that

$$(12) \quad \sum_{m=1}^{\infty} \sup_{0 < L < L_0} P[N(L)L^2/(\log \log L^{-2}) > m] < \infty$$

for some $L_0 > 0$. (See Bickel and Yahav (1968).) Now $P(N(L)L^2/(\log \log L^{-2}) > m)$

$$= P(N(L) > m(L)) \text{ where } m(L) = [m \log \log L^{-2}/L^2]$$

where $[x]$ denotes the largest integer $\leq x$.

$$\leq P(U_{m(L)} - V_{m(L)} > L)$$

$$= P(R_{m(L)} > L - \frac{2c(m(L), \alpha)}{m(L)D(F)}) \text{ from (6).}$$

Now there exist a positive integer M_0 and a positive number L_0 such that for $m \geq M_0$ and $L \leq L_0$ this probability is bounded from above by

$$P(R_{m(L)} > \lambda_1 L) \quad \text{where } \lambda_1 > 0$$

$$\leq P(R_{m(L)} > \lambda_2 (m(L))^{-1/2}) \quad \text{where } \lambda_2 > 0$$

$$\leq K'_s (m(L))^{-s} \text{ from lemma 1 (7)}$$

$$\leq K'_s m^{-s}.$$

Choosing $s > 1$ proves that (12) holds.

Asymptotic relative efficiency

Using methods analogous to the above, a $1-2\alpha$ confidence sequence of the form $\{\bar{X}_n \pm s_n a(n, \alpha)\}$ may be constructed for the mean μ of a normal distribution with unknown variance σ^2 where $\bar{X}_n$ and s_n are the sample mean and standard deviation based on n observations respectively and where $a(n, \alpha)$ is a sequence analogous to $c(n, \alpha)$:

Using the fact that $\sum_{i=1}^n X_i$ is a martingale when $\mu=0$, we apply Theorem A of the appendix to obtain a $1-\alpha$ confidence sequence for μ depending on σ . Also the fact that s_n^2 is a reverse martingale and a version of theorem A for reverse martingales leads to a $1-\alpha$ one-sided confidence sequence for σ . Combining these two confidence sequences, the above mentioned confidence sequence is obtained. The construction may be done in such a way that EN , the expected sample size of the associated bounded length confidence interval procedure, satisfies

$$EN/G(L) \rightarrow 1 \text{ as } L \rightarrow 0 \text{ where}$$

$$G(L) = 4\gamma B^2 \sigma^2 \log \log L^{-2}/L^2.$$

If the asymptotic relative efficiency of the nonparametric procedure with respect to this procedure is defined as the limit as $L \rightarrow 0$ of the ratio of expected sample sizes, we obtain the expression $4\sigma^2 D^2(F)/A^2$ which is the Pitman efficiency of a general rank order test with respect to the Student t-test.

5. Asymptotic theory as $\alpha \rightarrow 0$

Considering asymptotic theory as $\alpha \rightarrow 0$ should be of interest, since in practice α is small. We did not succeed in finding an asymptotic expression for EN , but the following theorem gives an asymptotic upper bound. The method used in the previous section is not applicable, since R_n in (6) depends on α .

Theorem 2

- (i) $N = N(\alpha)$ is a nondecreasing function of α as α decreases, $N \rightarrow \infty$ as $\alpha \rightarrow 0$ and $EN \rightarrow \infty$ as $\alpha \rightarrow 0$.
- (ii) $\limsup_{\alpha \rightarrow 0} N/h(\alpha) \leq 1$ a.s. where

$$h(\alpha) = \gamma A^2 (-\log \alpha) / 2 (\frac{1}{2}\mu - \rho(\frac{1}{2}L))^2, \text{ and}$$

$$\text{where } \rho(d) = \int_0^\infty J[H_{X-d}(x)] dF_{X-d}(x) \text{ and}$$

$$H_{X-d}(x) = P(|X-d| \leq x), F_{X-d}(x) = P(X-d \leq x).$$

(iii) The set $\{N/h(\alpha)\}_{\alpha > 0}$ is uniformly integrable and $\limsup_{\alpha \rightarrow 0} EN/h(\alpha) \leq 1$.

Proof:

(i) follows immediately from the definition of N and the Monotone Convergence Theorem.

(ii) Define two auxiliary stopping variables as follows:

$$R_1 = \min \{m \geq n_0 : U_n \leq \frac{1}{2}L \text{ for all } n \geq m\} \text{ and}$$

$$R_2 = \min \{m \geq n_0 : V_n \geq -\frac{1}{2}L \text{ for all } n \geq m\}.$$

Then $N \leq \max(R_1, R_2)$.

From the definition of R_1 we obtain

$$U_{R_1-1} > \frac{1}{2}L \text{ hence}$$

$$S_{R_1-1}(X_{R_1-1} - \frac{1}{2}L \mathbb{1}_{R_1-1}) \geq -c(R_1-1, \alpha).$$

Since $R_1 \rightarrow \infty$ a.s. as $\alpha \rightarrow 0$, we obtain

$$\limsup_{\alpha \rightarrow 0} R_1/c(R_1, \alpha) \leq (\frac{1}{2}\mu - \rho(\frac{1}{2}L))^{-1} \text{ a.s.}$$

This follows from Theorem 1 of Sen (1970). Using the fact (from (5)) that $b \rightarrow 2/B^2$ as $\alpha \rightarrow 0$ we obtain $\limsup_{\alpha \rightarrow 0} R_1/h(\alpha) \leq 1$ a.s. The same result holds for R_2 , which proves this part of the theorem.

(iii) Using the same method as in (iii) of Theorem 1 let $m(\alpha) = \lceil -m \log \alpha \rceil$. Then

$$P(N(\alpha)/(-\log \alpha) > m)$$

$$\leq P(U_{m(\alpha)} - V_{m(\alpha)} > L)$$

$$\leq P(U_{m(\alpha)} > \frac{1}{2}L) + P(V_{m(\alpha)} < -\frac{1}{2}L)$$

$$\leq P(S_{m(\alpha)}(X_{m(\alpha)} - \frac{1}{2}L \mathbb{1}_{m(\alpha)}) \geq -c(m(\alpha), \alpha))$$

$$+ P(S_{m(\alpha)}(X_{m(\alpha)} + \frac{1}{2}L \mathbb{1}_{m(\alpha)}) \leq c(m(\alpha), \alpha)).$$

The first term may be written as

$$(13) \quad P(T_{m(\alpha)}(X_{m(\alpha)} - \frac{1}{2}L_{m(\alpha)}) - \rho(\frac{1}{2}L) > t_{m(\alpha)})$$

$$\text{where } t_{m(\alpha)} = \frac{1}{2}\mu - c(m(\alpha), \alpha)/m(\alpha) - \rho(\frac{1}{2}L) > \delta (> 0)$$

for all $m \geq M_0$ and $\alpha < \alpha_0$ where M_0 and α_0 are appropriate positive constants. According to Theorem 2 of Sen (1970), this proves that (13) is bounded above by

$$2(\lambda(\delta))^{m(\alpha)} \text{ for } m(\alpha) \text{ large with } 0 < \lambda(\delta) < 1$$

$$\leq 2(\lambda(\delta))^m \text{ for } \alpha < \alpha_0 \text{ (taken small enough).}$$

A similar argument shows that this is also a bound for the second term above. This proves (iii).

Remark:

From the definitions of V_n and U_n we see that we must have $c(N, \alpha)/N \leq \frac{1}{2}\mu$ from which we obtain $\liminf_{\alpha \rightarrow 0} EN/(-\log \alpha) \geq 2\gamma$, so that asymptotically as $\alpha \rightarrow 0$, EN is between $2\gamma(-\log \alpha)$ and $h(\alpha)$.

APPENDIX

Theorem A

Let $\{Z_n\}$ be a martingale such that

$$(A1) \quad Ee^{aZ_n} \leq e^{\frac{1}{2}a^2n\tau^2}$$

where $\tau^2 > 0$, then for $c(n) = B\tau(\gamma n - 1)^{1/2}/(\log \log n - b \log \alpha)^{1/2}$ we have

$$P(Z_n \geq c(n) \text{ for some } n \geq m)$$

$$\leq \alpha^{\frac{1}{2}B^2b/(\frac{1}{2}B^2 - 1 + \log m/\log \gamma)}/(\frac{1}{2}B^2 - 1)(\log m)^{\frac{1}{2}B^2}$$

where $\gamma \geq 2$ and $m \geq 3$ are integers, $b > 0$ and $B > 2^{1/2}$ are constants and $0 < \alpha < 1$.

In order to prove the theorem we need the following lemma, the proof of which is due to J.H. Venter and J.W.H. Swanepoel.

Lemma

Let $\{d(n)\}$ and $\{a_i\}$ be sequences of constants, such that $\{d(n)\}$ is nondecreasing. Let $m = m_0, m_1, \dots$ be an increasing sequence of positive integers. Then

$$(A2) \quad P(Z_n \geq d(n) \text{ for some } n \geq m) \leq \sum_{i=0}^{\infty} e^{-a_i d(m_i)} E e^{a_i Z_{\ell_i}}$$

where $\ell_i = m_{i+1} - 1$.

Proof:

$$P(Z_n \geq d(n) \text{ for some } n \geq m)$$

$$= P\left(\bigcup_{i=0}^{\infty} (Z_n \geq d(n) \text{ for some } n \in [m_i, m_{i+1}])\right)$$

$$\leq \sum_{i=0}^{\infty} P(Z_n \geq d(m_i) \text{ for some } n \in [m_i, \ell_i]).$$

The lemma follows upon an application of a result of Steiger (1970) equation (5).

Proof of theorem A

Under (A1) the upper bound of the right-hand side of (A2) with $d(n) = c(n)$ is a minimum in a_i if $a_i = c(m_i)/\ell_i \tau^2$ ($i=1,2,\dots$). Hence the minimum value of the bound is

$$(A3) \quad \sum_{i=0}^{\infty} \exp \{-c^2(m_i)/2\ell_i \tau^2\}.$$

Let $m_i = m\gamma^i$, then since $\ell_i = \gamma m_i - 1$ (A3) becomes

$$\begin{aligned} & \sum_{i=0}^{\infty} \exp \{-\frac{1}{2} B^2 [\log(i \log \gamma + \log m) - b \log \alpha]\} \\ &= \alpha^{\frac{1}{2} B^2 b} (\log \gamma)^{-\frac{1}{2} B^2} \sum_{i=0}^{\infty} (i + \log m / \log \gamma)^{-\frac{1}{2} B^2}. \end{aligned}$$

This proves the theorem since

$$\sum_{i=0}^{\infty} (i + a)^{-t} \leq (t-1)^{-1} a^{-t} (a-1+t) \text{ for } t > 1.$$

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